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# Midterm Project Proposal:

# A Decade of Cyber Threats: Patterns, Risks and Vulnerabilities

## Background & Motivation

In recent years, cybersecurity threats have increased significantly, impacting businesses, governments, and individuals worldwide. With the rapid digital transformation, cyberattacks have become more sophisticated, leading to financial losses, data breaches, and operational disruptions. Analyzing historical cybersecurity threats can help identify patterns, trends, and risk factors that organizations must address to enhance their security posture.

## Importance of This Study

- Relevance – Cybersecurity is a growing concern for organizations, and understanding threat patterns can contribute to better security strategies.
- Data-Driven Insights – This dataset provides an opportunity to apply data science and SQL techniques to uncover key trends in cyber threats.
- Skill Development – This project allows me to gain hands-on experience with PostgreSQL, data analysis, and visualization while working with real-world cybersecurity data.

**Objective of the Project**

- To analyze global cybersecurity threat trends from 2015 to 2024.
- To identify common types of cyber threats and their frequency over time.
- To determine which industries or regions are most affected by cyberattacks.
- To derive actionable insights that can help businesses and policymakers mitigate cybersecurity risks.

**Dataset Description**

Our chosen dataset of Global Cybersecurity Threats includes 3000+ records, spanning from 2015 to 2024, with 10 columns/features. The columns use strings, integers, and decimals for the data collected. The features include the **Country** in which an attack took place (as a string), the **Year** the incident happened (as an integer), **Attack Type** of which cybersecurity threat occured (as a string), the **Target Industry** threatened (as a string), the **Financial Loss** in millions (as a decimal), the **Number of Affected Users** impacted from the attack (as an integer), **Attack Source** in which origin it occured (as a string), **Security Vulnerability Type** where the exploitation happened (as a string), the **Defense Mechanism Used** or strategy used against the attack (as a string), and the **Incident Resolution** of how long it took to resolve the attack (as an integer).

**Questions to Investigate**

- ❖ Trends and Patterns
  - ➢ How have cybersecurity threats evolved from 2015 to 2024?
  - ➢ What are the most common types of threats and how has that changed over time?

- ➢ Are there any seasonal trends we can look at?
    - Although this is interesting and is something we would like to look at, we unfortunately are not able to since the dataset only has years.

- ❖ Geographical Analysis
    - ➢ What countries are most affected by cyber threats?
        - Number of attacks, users affected, financial loss
    - ➢ Are certain types of attacks more common in some countries than others

- ❖ Industry Analysis
    - ➢ Which industries are most targeted?
    - ➢ Which industries have the biggest financial losses?

- ❖ Vulnerability and Defense Mechanism
    - ➢ What are the most exploited vulnerabilities?
    - ➢ What are the least effective defense mechanisms used?

- ❖ Response
    - ➢ What types of threats had the longest resolution times?

**Dataset Source**

https://www.kaggle.com/datasets/atharvasoundankar/global-cybersecurity-threats-2015-2024

# Database Screenshot

